$\Delta x \quad 10^{-6}$

29 **C** Alpha particles are stopped by paper,

Initial count of alpha emitter sample $= 352 - 256 = 96\ \text{min}^{-1}$
Initial count of beta emitter sample $= 256 - 16 = 240\ \text{min}^{-1}$

After 6 days, i.e. 144 h, alpha emitter undergoes 3 half-lives and beta emitter undergoes 4 half-lives.

Final count of alpha particle source $= \dfrac{96}{2^3} = 12\ \text{min}^{-1}$
Final count of beta particle source $= \dfrac{240}{2^4} = 15\ \text{min}^{-1}$

Total count $= 12 + 15 + 16 = 43\ \text{min}^{-1}$

30 **A** Energy released $= 5.607 \times 7 + 7.120 \times 4 - 6.494 \times 10 = 2.789\ \text{MeV}$